

Stacked Polyphase Bridge Converters for High-Voltage Electric Traction Drives: A Review of Topology, Modulation, Control, Semiconductor Scaling, and Future Directions

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ABSTRACT Increasing dc-link voltages in electric traction drives intensify semiconductor voltage stress and insulation requirements. Stacked polyphase bridge converters partition the dc-link voltage among series-connected three-phase bridges, each supplying an electrically separated machine winding group. This architecture enables lower-voltage semiconductor devices and modular converter-machine integration, but introduces coupled challenges in modulation, voltage balancing, insulation, and fault management. This paper reviews stacked polyphase bridge converters for high-voltage electric traction, bringing together literature on topology evolution, operating principles, machine integration, control, semiconductor scaling, and fault-tolerant reconfiguration. The review examines carrier-interleaving effects on dc-link ripple, conditions for switching-induced common-mode voltage cancellation, and dc-link balancing stability under centralized, common-duty-ratio, and distributed control. It also relates cell count to semiconductor voltage class and the associated converter and machine requirements. A mapping for a 1200-V dc link illustrates how increasing cell count provides access to lower-voltage device classes, while adding sensing, gate-drive, winding, and insulation requirements. The synthesis indicates that semiconductor-level gains alone are insufficient to establish a system-level advantage, and that the preferred cell count depends on the application and operating mission. Key research priorities are joint battery, converter, and machine design, together with validation against vehicle-level energy, cost, and reliability targets. These findings provide a basis for assessing where stacked polyphase bridge architectures can offer practical benefits in future traction drives.

INDEX TERMS Electric vehicles, integrated motor drives, modular converters, multiphase machines, stacked polyphase bridge (SPB) converters.

I. Introduction

Electrification is pushing traction-drive propulsion and fast-charging power toward several hundred kilowatts, while system-level performance requirements continue to rise [1]. Increasing the dc-link voltage has emerged as a primary design lever for meeting both demands. On the propulsion side, a higher dc-link voltage can improve efficiency and power density [2]. On the battery and charging side, it reduces the current required for a given power level, lowering cable conduction loss and conductor cross-section while enabling higher charging power under a given current limit [3].

This trend, however, is not without consequence at the converter level. A higher dc-link voltage also brings more severe insulation-coordination and electromagnetic-compatibility (EMC) demands, together with added thermal and reliability stress on the power stage [2], [3]. Raising the dc-link voltage therefore requires coordinated converter, machine, insulation, and thermal design.

Figure 1 summarizes these qualitative tradeoffs between conventional low-voltage and higher-voltage traction architectures.

TABLE 1. Representative voltage and power levels in published traction-inverter studies, industrial and reference platforms, and production battery-electric vehicles. Voltage classes and reported power ratings indicate application scale; vehicle totals, inverter ratings, and peak/continuous ratings are not directly comparable.

Published studies / demonstrators	Industrial / reference platforms	Production vehicles
Bertelshofer et al.: 400 V, 150 kW [6]	Bosch Gen. 4:400/800 V, 250/350 kW [7]	BMW i4 M50: 400 V, 400 kW [8]
Taha et al.: 800 V, 100 kW [9]	Wolfspeed reference design: 800 V, 300 kW [10]	Hyundai IONIQ 5 N: 800 V, 478 kW [11]
Al-Hmoud et al.: 1 kV, 200 kW [12]	Infineon eCAV platform: 800 V, 250 kW [13]	Volvo EX90 Performance: 800 V, 500 kW [14]
Wang et al.: 1.2 kV, 100 kW [15]	AVL PI850e: 800 V, 250 kW [16]	Porsche Taycan Turbo S: 800 V, 700 kW [17]
Ayaz et al.: 1.2 kV, 250 kW [18]	Punch IV5: 800 V, 600 kW [19]	Lucid Air Dream Performance: 900 V, 828 kW [20]

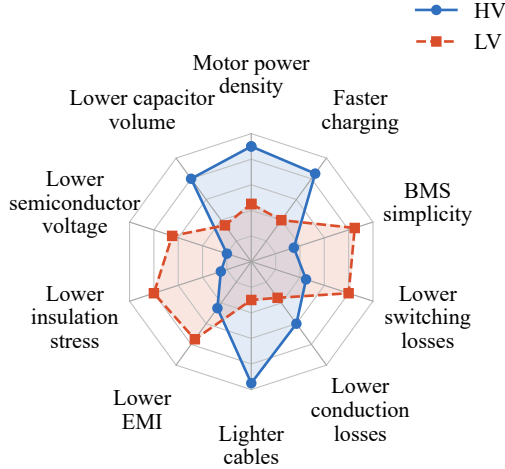


FIGURE 1. Conceptual illustration of qualitative system-level tradeoffs between low-voltage and high-voltage traction architectures. Radar scores indicate schematic relative tendencies, not measured values or a quantitative performance ranking.

A. Literature and Industry Trends Toward Higher DC-Link Voltage

The trend motivated above is not merely conceptual: it is already established in academic prototypes, industrial inverter platforms, and production vehicles, as summarized in Table 1. Reported academic demonstrators range from 400-V to 800-V and 1.2-kV converters, with still higher levels investigated for high-power passenger and heavy-duty applications [3], while industrial platforms increasingly target 800-V operation as a baseline and production vehicles show a comparable shift from established 400-V architectures to a growing number of 800-V and higher-voltage platforms [4], [5]. This industry-wide direction, rather than an isolated research interest, motivates the semiconductor-level analysis that follows since the rising dc-link voltages translate directly into a semiconductor-level bottleneck.

B. Semiconductor Blocking-Voltage Tradeoff

In a conventional two-level voltage-source inverter, each active switch must block a voltage directly tied to the complete dc-link voltage. In unipolar power devices, a higher blocking-voltage rating imposes a fundamental trade-

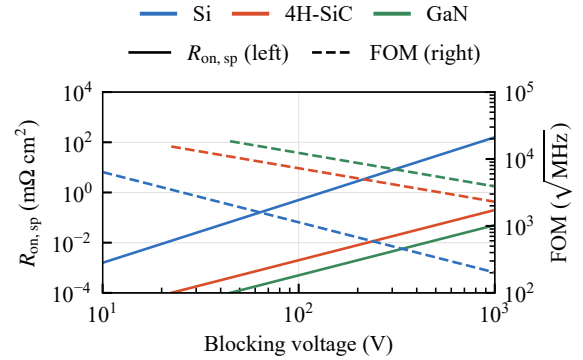


FIGURE 2. Ideal one-dimensional unipolar specific on-state resistance $R_{on,sp}$ (solid, left axis) and the resulting high-frequency figure of merit $FOM = 1/\sqrt{R_{on,sp}C_{oss,sp}}$ (dashed, right axis), in $\sqrt{\text{MHz}}$ – the product $R_{on,sp}C_{oss,sp}$ is an area-independent RC time constant, versus blocking voltage for Si, 4H-SiC, and GaN.

off against both specific on-state resistance and switching performance, commonly captured by the figure of merit, as shown in Fig. 2 [21], [22]. This ideal scaling favors lower blocking voltage within a given material model; commercial-device comparisons must also account for structure, temperature, packaging, and switching conditions.

Wide-bandgap (WBG) semiconductors shift this FOM favorably: SiC metal-oxide-semiconductor field-effect transistors (MOSFETs) now dominate high-voltage automotive drives, while GaN is particularly attractive at lower voltage owing to its fast switching behavior and superior FOM [21], [23]–[25]. Yet switching material does not remove the underlying coupling between dc-link voltage and device voltage class: a higher dc-link voltage can require a higher device voltage class, with conduction and switching losses determined by the selected technology and operating conditions. Converter architectures that partition the dc-link voltage among multiple switching units, so each device blocks only a fraction of the total voltage, therefore become increasingly relevant as system voltage rises.

C. Multilevel Converters and Their Limitations

Multilevel converters constitute an established solution to this voltage-partitioning problem. NPC, T-type, flying-capacitor, and cascaded architectures all distribute the dc-

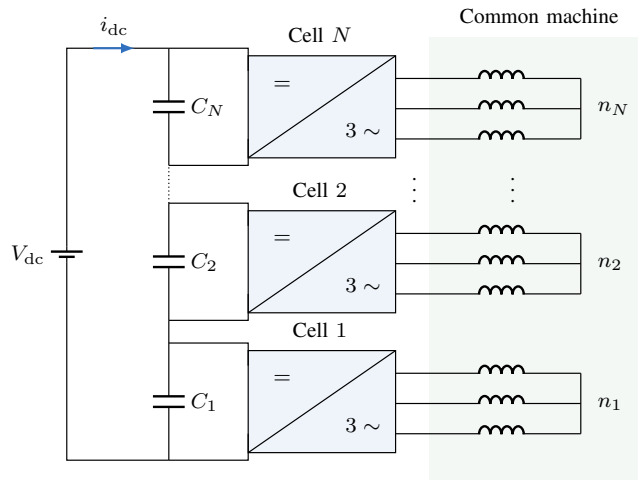


FIGURE 3. Conceptual architecture of an N -cell SPB traction drive: series-connected local dc-link capacitors, series-connected three-phase two-level bridge cells, and their electrically separated machine winding groups.

link voltage among additional semiconductor positions or capacitor-voltage levels, reducing individual device voltage stress [26]–[29].

Voltage partitioning requires additional semiconductor positions, passive components, and gate drivers. Classical three-phase multilevel bridges still supply a common winding set, whereas cascaded and other modular architectures can provide module-level redundancy. In either case, fault tolerance depends on available isolation paths, spare voltage capability, and control; component count alone establishes neither fault tolerance nor system reliability.

D. Stacked Polyphase Bridge

The stacked polyphase bridge (SPB) distributes the high-voltage conversion function among N repeated converter units. Each unit is a conventional three-phase two-level bridge connected in series on the dc side, supplying its own electrically separated three-phase winding group of the same machine, as illustrated in Fig. 3.

Under balanced operation, each cell's local dc-link voltage $V_{dc,k} \approx V_{dc}/N$, so the semiconductor voltage rating can be selected on the local cell voltage rather than the complete traction dc-link voltage – addressing the semiconductor bottleneck much as multilevel converters do. Each cell also drives a separated winding group, providing modularity at the converter–machine interface. A faulted cell can, in principle, be isolated and bypassed while the remaining cells continue to drive the machine at reduced power. It is this combination of semiconductor voltage reduction and converter–machine modularity that distinguishes SPB from conventional multilevel and cascaded architectures and motivates its consideration as a candidate topology for next-generation traction drives.

Realizing this potential, however, introduces coupled design challenges in component count, dc-link dynamics, ma-

chine design, common-mode voltage, and fault handling, addressed throughout this review. These challenges have grown more tractable as neighboring research areas have matured: multiphase machines provide additional post-fault degrees of freedom, integrated modular motor drives show how repeated power-electronic units can be physically associated with individual machine sections, wide-bandgap semiconductors amplify performance differences among voltage classes, and distributed control coordinates modular converter units [30]–[35].

Despite more than a decade of SPB research, beginning with Norrga *et al.* [36], the feasibility of electric-vehicle traction applications requires these developments to be assessed together. The available literature remains distributed across converter topology, modulation, dc-link stability, multiphase-machine design, integrated motor drives, semiconductor technology, thermal management, distributed control, and fault-tolerant operation. Although these topics are strongly coupled in an SPB traction drive, they are rarely evaluated within a common system boundary, leaving a gap between a well-motivated architectural concept and a system-level feasibility assessment. Accordingly, this paper provides a system-level review of SPB traction-drive technology aimed at closing that gap.

E. Review Scope and Contribution

This is a focused narrative review rather than an exhaustive systematic survey or a meta-analysis. Direct SPB evidence concerns series-connected dc ports supplying electrically separated winding groups; related multiphase and integrated-drive studies provide enabling methods but are not treated as demonstrations of SPB traction performance. The review distinguishes reported results, analytical illustrations, and application-level inferences. Its contribution is to connect device voltage partitioning to winding feasibility, modulation, balancing, common-mode excitation, and post-fault operation within one EV design boundary. Numerical examples illustrate these dependencies and do not constitute automotive validation.

The remainder of the paper is organized as follows. First, SPB's historical development, operating principle, and modulation are consolidated. Second, the interaction between cell count and the electric machine is examined alongside multiphase-machine and integrated-modular-drive research, including the insulation-coordination overhead of series-connected cells. Third, common-mode voltage and dc-link dynamics and balancing are examined. Fourth, semiconductor voltage scaling and fault handling are treated as coupled architecture-level design problems. Finally, the remaining research gaps and automotive validation requirements are identified.

II. Evolution of the Stacked Polyphase Bridge Concept

SPB belongs to the broader development of modular converter–machine systems. Modular and integrated modular

motor drives established repeated converter–winding units, machine segmentation, power sharing, and fault-tolerant operation [30], [37]. Series-connected implementations extended these concepts to dc-voltage sharing, enabling lower-voltage semiconductor devices [31], [38], [39]. Other modular drive topologies adopted the same principle [40], [41].

Norrgra *et al.* introduced SPB in 2013 [36], followed by dedicated control and modulation methods [42]. Related series-connected drives demonstrated internal dc-link regulation using machine-side degrees of freedom [43]. Subsequent stability analyses established the balancing-controller requirements [44], [45], identifying dc-link balancing as a central control challenge alongside semiconductor voltage selection.

Research then expanded toward coordinated modulation, scalable control, and converter–machine design. Modulation studies linked switching loss to dc-link harmonics [46], while distributed control provided an alternative to centralized balancing [47]. Comparative assessment against parallel-connected polyphase bridges and other modular converters evaluated machine design, losses, capacitor energy, cost, and redundancy, positioning SPB among several candidate architectures [48]. Interleaving and phase-count studies further connected converter performance to machine architecture [33], [49]–[51].

Recent work has extended SPB toward decentralized and reconfigurable operation. Common-duty-ratio control, decentralized balancing, and multi-agent power distribution reduce reliance on centralized coordination [52]–[55]. Online module isolation and reintroduction enable dynamic dc-bus reconfiguration, including removal of a faulted cell without interrupting torque production [56]. In parallel, gallium nitride devices have renewed interest in the benefits of lower semiconductor voltage classes [57].

Fig. 4 groups these developments by research focus, tracing the progression from voltage partitioning to coordinated converter–machine design and reconfigurable operation. The remaining integration gap for electric vehicles is a unified, application-specific SPB design framework that couples battery and dc-link requirements, semiconductor selection, machine configuration, and control, evaluated against common energy, power-density, cost, and reliability targets.

III. Operating Principle and Topological Characteristics

A. Basic Structure and Voltage Partitioning

An N -cell SPB consists of N conventional three-phase two-level voltage-source inverter cells connected in series on the dc side, with each bridge supplying one electrically separated three-phase winding group of the same machine [36], [42], [46]. The machine therefore presents N ac ports, while the converter cells share a single series dc path.

The dc-link voltage satisfies

$$V_{dc} = \sum_{k=1}^N V_{dc,k}, \quad V_{dc,k} \approx \frac{V_{dc}}{N} \quad (1)$$

2013–2014 Topology and control

SPB introduction and dedicated control and modulation [36], [42].

2015–2017 Stability and architectural assessment

Dc-link stability and balancing requirements [44], [45]; comparison with PPB-I/II and MHF candidates [48].

2017–2018 Modulation and distributed control

Switching-loss and dc-link harmonic shaping [46]; distributed balancing [47].

2019–2021 Converter–machine coordination

Interleaving, dc-link ripple, and machine phase-count selection [33], [49]–[51].

2023–2024 Reconfiguration and device integration

Common-duty-ratio and decentralized balancing, power sharing, and gate-delay effects [52]–[55]; online module reconfiguration [56]; 650-V GaN SPB comparison with a three-level flying-capacitor converter [57].

Remaining integration gap for EVs

Open research direction

Unified SPB design for EV applications

Joint selection of cell count, semiconductor technology, winding configuration, modulation, and cooling under battery/dc-link constraints and vehicle operating profiles. Multi-objective evaluation of energy consumption, power density, cost, and reliability.

FIGURE 4. Representative developments in SPB research grouped by period and research focus. Overlapping periods reflect concurrent progress in stability, modulation, control, and converter–machine integration. The dashed block identifies the EV-specific research direction synthesized in this review.

under balanced operation. A basic implementation contains $6N$ active switches and N local dc-link capacitors. Increasing N reduces the nominal cell voltage, and hence the required semiconductor voltage class, but simultaneously increases the number of switches, gate drivers, sensors, capacitors, and machine-winding terminals that must be realized and managed. Fig. 5 shows this structure at switch level for two representative cells, making explicit the $6N$ -switch bridge realization, the local capacitors C_k , and the notation used throughout the remainder of this paper.

B. Series DC Connection and Power Flow

Neglecting auxiliary and bypass paths, the cell dc currents are constrained to be equal,

$$i_{dc,1} = i_{dc,2} = \dots = i_{dc,N} = i_{dc}, \quad (2)$$

so that the dc-side power entering cell k is

$$p_{dc,k} = V_{dc,k} i_{dc}. \quad (3)$$

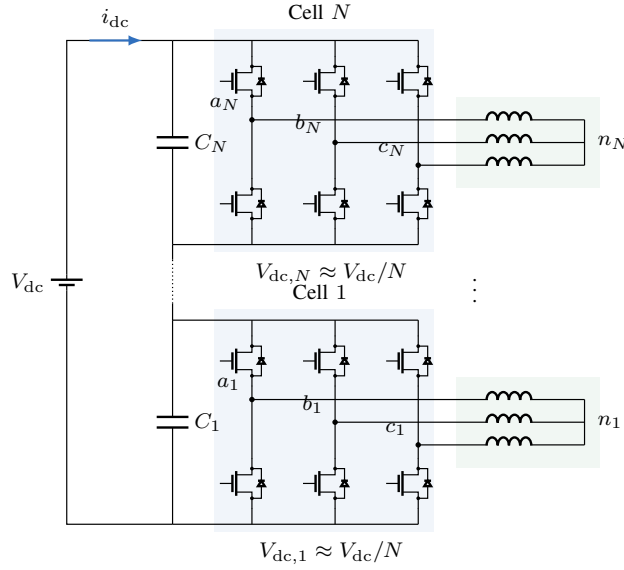


FIGURE 5. Switch-level realization of the SPB dc-ac stage for two representative cells (Cell 1 and Cell N ; cells $2, \dots, N-1$ elided). The common series current i_{dc} (2) flows through every cell, giving each cell the dc-side power $p_{dc,k} = V_{dc,k} i_{dc}$ (3).

The total machine power, by contrast, is the sum of the powers delivered to the individual winding groups,

$$P_{ac} = \sum_{k=1}^N P_k. \quad (4)$$

Under symmetrical operation $P_k \approx P_{ac}/N$, but equal power sharing is not a fundamental requirement of the topology. The multi-three-phase machine provides additional current and power degrees of freedom that allow power to be redistributed among winding groups [43], [54]. This redistribution capability is the physical basis for both dc-link voltage balancing and several of the post-fault operating strategies.

The combination of a common dc current with independently controllable ac-side powers also explains why the local capacitor voltages must be treated as internal dynamic states rather than fixed design parameters. If one winding group processes more power than another, the corresponding capacitor energies can diverge even though all cells carry the same dc current [44], [45]. Managing this divergence is the dc-link voltage-balancing problem, SPB's first-order control challenge.

C. Cell Count as a Converter-Machine Design Variable

Cell count is consequently a converter-machine design variable rather than a semiconductor variable alone. Increasing N also reduces the voltage available to each winding group and alters the number of winding sets, terminal count, electromagnetic coupling, current-control degrees of freedom, and available reconfiguration states.

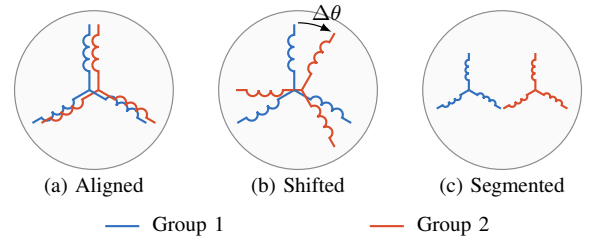


FIGURE 6. Schematic comparison of winding-group spatial arrangements, each coil marking one phase axis of winding group 1 (blue) or group 2 (red): (a) aligned groups (parallel phase axes), (b) spatially shifted groups (offset by $\Delta\theta$), and (c) physically segmented stator modules.

Machine phase current, however, does not generally scale as $1/N$: it instead depends on the number of turns, flux linkage, winding connection, torque constant, and machine geometry. A many-cell architecture may therefore reduce the semiconductor voltage class while still requiring high-current devices or parallel chips at the device level – a coupling among cell count, voltage class, and current rating that matters most as the design space extends toward lower-voltage device classes.

IV. Electric-Machine Requirements and Integrated Implementation

A. Winding Architecture and Electromagnetic Coupling

Although the winding groups are electrically separated at their terminals, they share the same magnetic circuit, rotor, air gap, and shaft. Their electromagnetic coupling depends on winding distribution, slot/pole combination, spatial displacement, and machine geometry [33], [43], [54], [58]. Therefore, an SPB-compatible machine cannot generally be obtained simply by dividing a conventional three-phase winding into multiple groups without considering the resulting coupling and harmonic behavior.

The three-phase groups may be aligned, spatially shifted, or implemented as physically segmented stator modules. Spatial displacement affects mutual coupling, torque-ripple components, and the additional current subspaces available to the controller, among other effects [59]–[61]. These additional subspaces can introduce non-torque-producing currents, but they can also be exploited for harmonic suppression, differential power sharing, and fault-tolerant operation [54], [62]. Split winding architectures show a similar tradeoff: they can reduce ripple and provide additional current/voltage degrees of freedom, but they do so by increasing winding terminals and coupling complexity [63]. These three winding-group arrangements are compared schematically in Fig. 6.

For the k th PMSM winding group, a steady-state approximation neglecting mutual coupling between groups gives

$$\begin{aligned} v_{d,k} &= R_{s,k} i_{d,k} - \omega_e L_{q,k} i_{q,k}, \\ v_{q,k} &= R_{s,k} i_{q,k} + \omega_e (L_{d,k} i_{d,k} + \psi_{f,k}), \end{aligned} \quad (5)$$

For coupled or transient operation, the flux-linkage model must retain $\dot{\psi}_{d,k}$ and $\dot{\psi}_{q,k}$ and the dependence of flux on all winding-group currents. The required voltage must satisfy

$$\sqrt{v_{d,k}^2 + v_{q,k}^2} \leq k_{\text{mod}} V_{\text{dc},k} \approx k_{\text{mod}} \frac{V_{\text{dc}}}{N}. \quad (6)$$

This falling per-cell voltage budget ties winding turns, inductance, and field-weakening capability directly to N – a coupling demonstrated in optimized multiphase SPB-PMSM systems [33]. Recent multiphase traction-machine studies similarly emphasize coordinated design of winding voltage, harmonic subspaces, and modulation [64], [65].

B. From Multiphase Machine to Integrated Modular Drive

Physical integration can convert the added electrical complexity of many winding groups into a packaging advantage. With a remote inverter, additional winding groups increase phase-cable and connector count; positioning each converter bridge close to its corresponding winding group instead reduces external cabling, shortens high-frequency interconnections, distributes heat sources, and aligns converter modules with stator segments.

This design philosophy predates SPB and has been extensively studied in integrated and modular motor drives [30], [31], [34], [38], [39], [66]–[78], summarized in Table 2. The overall trend runs from early cable-reduction and compact-packaging concepts through jointly designed fault-tolerant converter/winding/thermal architectures and series-connected GaN IMMDs to recent WBG-based axial-flux IMMDs demonstrating stator-mounted converter modules and scalability to MW-class and EV-traction implementations.

C. Physical Integration Options

The electrically separated winding groups required by SPB do not impose a unique converter placement on a radial-flux machine. Chowdhury *et al.* [79], consistent with the broader integrated-motor-drive survey of Abebe *et al.* [80], classify motor/inverter physical integration into four categories: whether the converter mounts radially or axially, and whether it interfaces with a separate housing/end-cap structure or directly with the stator core. This taxonomy frames how SPB converter cells can be placed around a radial-flux machine, as illustrated in Fig. 7. The axial-flux stator-mounted and polygon-housing alternative is covered by the Mohamed *et al.* developments [74]–[76] in Table 2.

Modern WBG drives therefore provide several machine and packaging concepts that are compatible with SPB, including segmented windings, distributed converter cells, shared cooling, and integrated dc-link structures. These developments do not establish an intrinsic SPB advantage, but they demonstrate the availability of enabling technologies required for practical implementation.

D. Integration Is Not Free of Complexity

Integration does not remove complexity; it relocates it. Converter modules mounted on or near the machine are ex-

TABLE 2. Representative Integrated Motor-Drive Developments

Reference	Integration theme
Shakweh <i>et al.</i> [66]	Integrated motor/inverter architecture
Wheeler/Pickering <i>et al.</i> [67], [68]	Converter and thermal design inside motor envelope
Brown <i>et al.</i> [69]	Converter mounted at modular stator pole
Wolmarans <i>et al.</i> [70]	Fault-tolerant converter, machine, and thermal design
Choi <i>et al.</i> [30]	Modular PM machine and high-temperature electronics
Wang <i>et al.</i> [38], [39]	Low-voltage GaN, dc-voltage sharing, and interleaving
Uğur and Keysan [72], [73]	Multi-physics converter-machine optimization
Mohamed <i>et al.</i> [74]–[76]	Stator-mounted modules, polygon housing, integrated dc link
Wang <i>et al.</i> [77]	Repeated machine and converter submodules
Chen <i>et al.</i> [78]	WBG converter, machine, passive, and cooling co-design
Deng <i>et al.</i> [34]	Recent IMD technologies and research directions

posed to elevated temperature, vibration, and sealing/coolant requirements, among other stresses, and to electromagnetic coupling between gate-drive, power, and machine conductors. Shared machine-converter cooling can also increase thermal interaction between the stator and power electronics.

SPB introduces an additional insulation challenge because its converter cells and associated winding groups occupy different absolute potentials along the series dc stack. Gate-driver supplies, sensors, and mechanical mounting, among other subsystems, must therefore preserve the required electrical isolation.

Consequently, practical SPB integration requires coordinated electrical, thermal, and mechanical design, complementing the semiconductor- and system-level tradeoffs.

V. Modulation Strategies

Each stacked polyphase bridge (SPB) cell supports conventional sinusoidal pulsewidth modulation (SPWM), space-vector modulation (SVM), and discontinuous pulsewidth modulation (DPWM). However, the shared dc current, dynamic cell voltages, and electromagnetic coupling between winding groups require coordination across cells. Carrier phase, zero-sequence injection, switching sequence, and clamping intervals provide additional degrees of freedom for shaping dc-link ripple, machine harmonics, switching loss, common-mode voltage (CMV), and winding-group power distribution [42], [46]. Table 3 groups these degrees of freedom by their principal system-level effects.

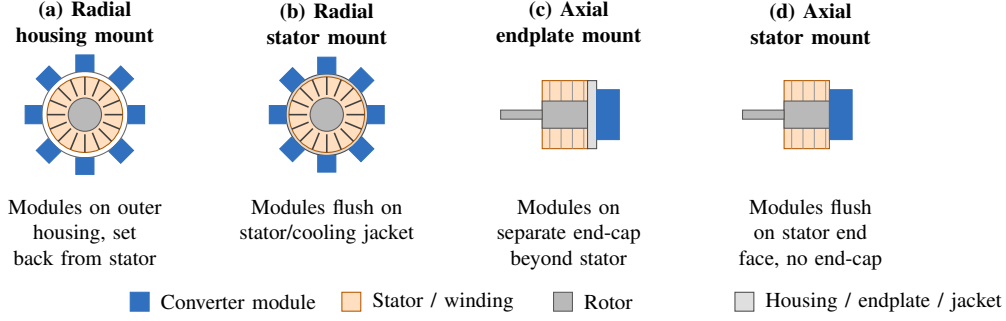


FIGURE 7. Physical integration options for SPB-compatible converter modules on a radial-flux machine, following the four-category taxonomy of [79]: (a) radial housing mount, (b) radial stator mount, (c) axial endplate mount, and (d) axial stator mount, each shown in whichever radial or axial cross-section most clearly conveys its mounting geometry.

TABLE 3. SPB modulation degrees of freedom grouped by principal system-level effect.

Effect	Degrees of freedom
Ripple and loss shaping	Carrier phase [46], [51], [81]; switching sequence [82], [83]; clamping and DPWM [84], [85]
CMV and balancing authority	Zero-sequence injection and vector redundancy [86], [87]; differential cell commands [54], [88]

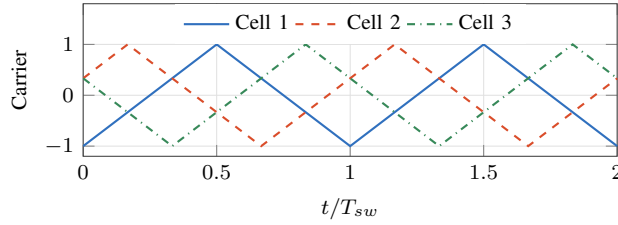


FIGURE 8. Uniform carrier interleaving for $N = 3$, with phase offsets 0 , $2\pi/3$, and $4\pi/3$. Analytically generated triangular carriers.

A. Carrier Coordination and Harmonic Cancellation

For a binary upper-switch command $s_{x,k} \in \{0,1\}$, the double-Fourier representation is

$$s_{x,k}(t) = \sum_{m,n \in \mathbb{Z}} c_{mn,x,k} e^{j[m(\omega_c t + \theta_{c,k}) + n(\omega_0 t + \theta_{0,k})]}. \quad (7)$$

Here m and n are carrier and fundamental indices. The coefficients include the dc term, baseband harmonics, carrier harmonics, and sidebands; conjugate symmetry gives a real switching function. A common carrier frequency and fundamental frequency are assumed across cells.

The carrier phase is a free parameter for shaping aggregate harmonics. Uniform interleaving assigns

$$\theta_{c,k} = \theta_{c,1} + (k-1)\Delta\theta_c, \quad \Delta\theta_c = \frac{2\pi}{N}. \quad (8)$$

Fig. 8 illustrates this arrangement for three cells.

B. DC-Link Ripple and Battery-Side Current

DC-link current ripple affects capacitor loading and the current presented to the battery and input filter. Higher

switching frequency can reduce passive-component requirements within a specified loss budget, while interleaving shapes the aggregate spectrum.

With phase current positive from converter to machine, the ideal cell input current is $i_{\text{conv},k} = \sum_x s_{x,k} i_{x,k}$. Consequently, its spectrum is

$$I_{\text{conv},k} = \sum_{x \in \{a,b,c\}} S_{x,k} \otimes I_{x,k}. \quad (9)$$

For matched cell spectra with a common fundamental phase, uniform interleaving multiplies carrier-order m by

$$\sum_{k=1}^N e^{jm2\pi(k-1)/N} = \begin{cases} N, & m \in N\mathbb{Z}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

This selection rule cancels nonmultiples of N ; it does not guarantee a nonzero component exactly at Nf_{sw} . The first surviving term also depends on the switching coefficients, sideband index, and winding/current phase relationships [42]. Unequal cell loading or voltage sharing weakens cancellation.

The shared dc-link current follows from the equivalent circuit developed in Section VII:

$$L_b \frac{di_{\text{dc}}}{dt} = E_b - R_b i_{\text{dc}} - \sum_k v_{\text{conv},k}, \quad (11)$$

$$C_k \frac{dv_{\text{conv},k}}{dt} = i_{\text{dc}} - i_{\text{conv},k}.$$

For equal cell capacitances, $C_k = C$, the relationship between the summed converter-current spectrum and dc-bus current is [42]

$$I_{\text{dc}}(\omega) = \frac{\sum_{k=1}^N I_{\text{conv},k}(\omega)}{N - \omega^2 L_b C + j\omega C R_b}. \quad (12)$$

Equation (12) assumes identical capacitors, a fixed source voltage, and prescribed converter-current spectra. For positive-frequency peak phasors, a consistent ripple measure is

$$I_{\text{dc,rms}}^{\text{ripple}} = \left[\frac{1}{2} \sum_{\omega_h > 0} \frac{|\sum_k I_{\text{conv},k}(\omega_h)|^2}{|N - \omega_h^2 L_b C + j\omega_h C R_b|^2} \right]^{1/2}. \quad (13)$$

Contributions at the same frequency are combined before taking the magnitude. Retained harmonics must be evaluated

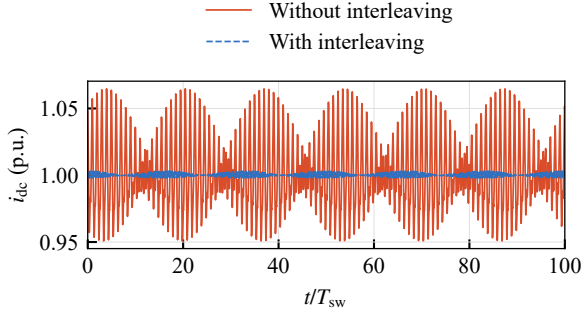


FIGURE 9. Analytical steady-state shared dc-link current for three identical SPB cells with ideal natural SPWM over one fundamental period. Parameters are $M = 0.85$, power-factor angle 20° , $f_{sw}/f_1 = 100$, $L_b C/T_{sw}^2 = 0.85$, and $R_b C/T_{sw} = 2.5$. Exact switching-interval Fourier integrals are propagated through (12), retaining 8000 fundamental harmonics. Both averages are 1 p.u.; ripple RMS is 0.0281 p.u. without interleaving and 0.00126 p.u. with uniform interleaving. These are ideal-model predictions, not measurements.

against the input-network resonance; interleaving does not ensure attenuation at every frequency. Fig. 9 illustrates one operating point.

The analytical example includes lumped battery-to-stack inductance but assumes identical cells, prescribed sinusoidal phase currents, ideal switching, and ideal cell connections. Periodic steady-state reconstruction excludes startup transients; doubling the harmonic cutoff from 4000 to 8000 changes the current by less than 1.3×10^{-5} p.u. In distributed implementations, bus-bar and connector-loop inductances alter harmonic cancellation. A series-connected gallium nitride integrated modular motor drive required careful capacitor selection and compact layout to preserve interleaving benefits [39]; an earlier characterization found that parasitic loop inductance could make interleaved ripple exceed the noninterleaved value [89]. The illustrated reduction should therefore be treated as case-specific: achievable performance also depends on layout, capacitor selection, and switching frequency.

C. Machine Harmonics, Voltage Utilization, and Switching Loss

The preferred carrier displacement depends on cell count, modulation index, power factor, winding displacement, and the optimization objective. SPB studies demonstrate its influence on semiconductor losses and dc-link current harmonics [46], while related modular and multiphase drives demonstrate simultaneous reductions in dc-link and machine current ripple [51], [82], [90].

In multiphase machines, nonfundamental voltages can excite low-impedance harmonic subspaces, increasing copper loss and torque ripple despite satisfactory fundamental current control. Interleaved and discontinuous modulation strategies therefore require adaptation to the multiphase current-ripple characteristics [81], [91], [92].

At high speed, voltage utilization becomes critical because each winding group is supplied by its local cell voltage $V_{dc,k}$, subject to the machine voltage limit in (6). This constraint becomes more restrictive as cell count increases and local voltage decreases. Optimized vector selection in dual-three-phase traction machines can extend operation in the high-modulation region while suppressing unwanted subspace currents [64].

Switching-loss reduction introduces a competing objective. DPWM and hybrid strategies reduce transitions by clamping selected phases or redistributing high-frequency switching among modules. Related open-end-winding studies demonstrate simultaneous reductions in switching loss and zero-sequence current [84], [85]. Although their dc architecture differs, the principle of coordinating switching activity among inverter modules is applicable to SPB.

D. Coupling to Voltage Balancing and Common-Mode Control

Modulation also determines the control authority available for dc-link voltage balancing and CMV reduction. Voltage balancing requires differential winding-group power commands that remain feasible within local voltage and current limits [54], [55], [88]. Redundant voltage vectors and coordinated zero-sequence injection can reduce common-mode excitation of the machine frame [86], [87], [93]. These objectives are examined in Sections VII and VI, respectively.

SPB modulation consequently requires coordinated switching decisions that account for ripple, losses, voltage utilization, balancing, and common-mode behavior across the operating map.

VI. Common-Mode Voltage Effects

In any converter-fed ac machine, the non-zero average of the three phase-leg pole voltages produced by PWM switching gives rise to a common-mode voltage (CMV) that excites parasitic capacitive paths between winding, frame, shaft, and bearings. This mechanism is well characterized for a converter with a single machine port whose voltage scaling is produced internally, such as the flying-capacitor multilevel converter, for which an analytical CM-EMI model referenced to one dc-link potential exists [94]. SPB does not fit this picture: because its N cells are connected in series on the dc side (Section III) while each feeds an electrically separate winding group, cell k 's local dc rails are displaced from the overall dc-bus reference by an amount that depends on its position in the stack.

A. Series-Stack Origin of Common-Mode Excitation

Taking cell 1's negative dc rail as the reference, with cell k 's negative rail coinciding with cell $(k-1)$'s positive rail, the absolute offset of cell k is

$$V_{-,k} = \sum_{j=1}^{k-1} V_{dc,j} \approx (k-1) \frac{V_{dc}}{N}, \quad (14)$$

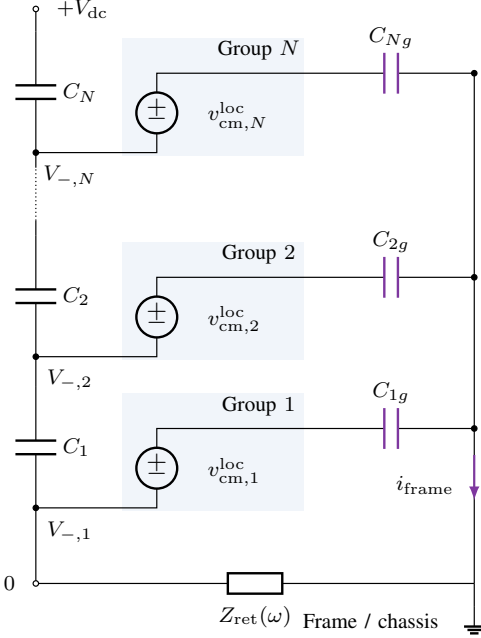


FIGURE 10. Lumped common-mode equivalent of the SPB stack. Series dc-link capacitors C_k establish the rail offsets $V_{-,k}$; each voltage source represents $v_{cm,k}^{loc} = (v_{aN_k} + v_{bN_k} + v_{cN_k})/3$. The group-to-frame capacitance C_{kg} couples this local source and its rail offset to the chassis. $Z_{ret}(\omega)$ represents the installation-dependent return path to the dc reference, not an assumed galvanic bond. Differential-mode winding circuits and the rotor/bearing network are omitted.

using the voltage-partitioning relation (1). The common-mode potential seen by winding group k is then

$$v_{cm,k}^{abs}(t) = V_{-,k} + \frac{1}{3} (v_{aN_k} + v_{bN_k} + v_{cN_k}), \quad (15)$$

where $v_{aN_k}, v_{bN_k}, v_{cN_k}$ are cell k 's pole voltages referenced to its own negative rail N_k . The second term is the conventional switching-induced CMV already addressed by zero-sequence injection and vector-redundancy modulation (Section V); the first term, $V_{-,k}$, is specific to SPB, set by the cell's stack position and by the instantaneous voltage-sharing state (Section VII), not by modulation alone.

Fig. 10 illustrates the resulting structure: increasing absolute potential offset with stack position, together with representative CM current paths coupling each winding group to the shared frame and chassis ground through parasitic capacitance. Because the cells and their winding groups are physically close together in the integrated implementations of Section IV, these paths are short and the coupling between cells is direct rather than mediated by long cabling. Physically integrated motor-drive assemblies show that common-mode current can resonate through exactly this kind of parasitic loop, and that the resonance is not adequately predicted by a single-source CM model [93], [95] – extending such modeling to a multi-port structure with N distinct dc offsets is, to the authors' knowledge, not yet available for SPB specifically.

B. Carrier Synchronization for CMV Cancellation

Beyond the stack-offset mechanism above, synchronizing each cell's PWM waveform relative to the others is a way to actively cancel the switching-induced component of CMV. We illustrate the principle for the $N = 2$ case, where the cancellation condition has a simple closed form; the same synchronization idea extends to larger stacks built from matched cell pairs, subject to matching and the required winding arrangement.

For equal cell voltages V_c , define the midpoint-referenced term $u_{cm,k} = V_c(\sum_x s_{x,k}/3 - 1/2)$. Complementary commands $s_{x,2} = 1 - s_{x,1}$ give $u_{cm,2} = -u_{cm,1}$; reversing the second winding group's sense preserves the intended torque direction. The absolute voltages remain offset:

$$\frac{v_{cm,1}^{abs} + v_{cm,2}^{abs}}{2} = V_c = \frac{V_{dc}}{2}. \quad (16)$$

Thus switching ripple cancels in the arithmetic mean, not the dc potential relative to the bottom rail. This This balanced-cancellation principle was first demonstrated for a dual-winding-set machine driven by a balanced inverter pair on a single, non-series-stacked machine [96], and was later applied to a genuine two-cell series-stacked (2L-SSC) SPB inverter [57]; as with interleaved GaN IMMDs, the achievable reduction is ultimately limited by layout parasitics and local dc-link capacitor design [39].

This cancellation assumes perfectly matched, simultaneous switching transitions between the two cells: a propagation-delay mismatch (e.g., from PCB layout or driver differences) reintroduces a residual CM spike, of duration equal to the delay, at every misaligned transition, and unequal power sharing during dc-link balancing similarly degrades the cancellation, both shown quantitatively in [57], [96]. The mechanism is specific to $N = 2$, or to a stack of matched cell pairs, so it should be read as a valuable special-case mitigation rather than a general solution to the stack-offset problem above. Fig. 11 illustrates the reference levels. For fixed lumped winding-to-frame capacitances C_{kg} , the frame excitation is approximately

$$i_{frame} = \sum_k C_{kg} \frac{d}{dt} (v_{cm,k}^{abs} - v_{frame}). \quad (17)$$

Cancellation of the arithmetic mean suppresses this current only for matched coupling paths. Unequal capacitances, switching delays, and resonances leave residual excitation. A constant stack offset contributes insulation stress but no steady capacitive current. Bearing current requires the additional winding–rotor–shaft–bearing network and cannot be inferred directly from this lumped frame-current relation.

C. Physical Consequences and Mitigation

CMV-driven bearing current is a documented failure mechanism in inverter-fed machines generally, arising from capacitive coupling through the air gap and discharge events once the bearing lubricant film breaks down [97], and can be identified with dedicated non-intrusive current-measurement

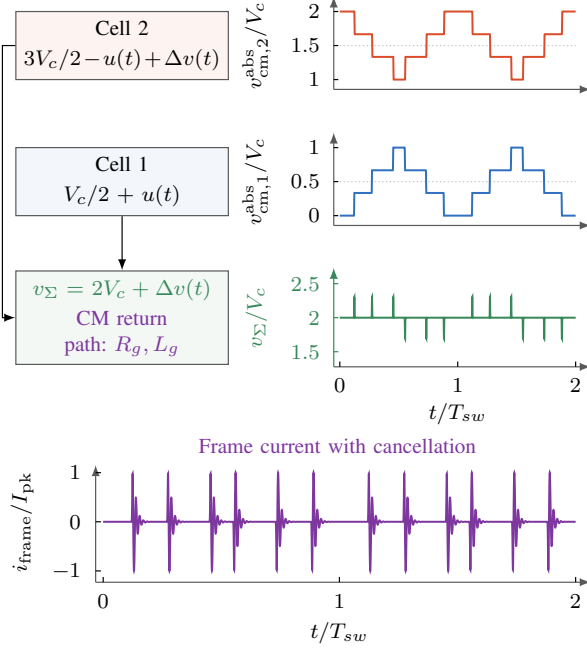


FIGURE 11. Nonideal two-cell common-mode cancellation over two switching periods. Small transition misalignment produces narrow residuals in v_{Σ} ($t_d/T_{sw} = 0.003$, $t_r/T_{sw} = 0.004$). Their polarities follow the three rising and three falling cell-voltage transitions; unequal dv/dt can also produce residuals. The lower plot shows damped frame current with cancellation, normalized to its own peak I_{pk} . For equal winding-to-frame capacitances C_g and a shared frame-to-ground return impedance $R_g + sL_g$, $I_{frame}(s)/V_{\Sigma}(s) = sC_g/(1 + 2R_gC_g s + 2L_gC_g s^2)$. The illustrative resonance has $f_0/f_{sw} = 65$ and damping ratio $\zeta = 0.22$. Each residual excites alternating-polarity current ringing. This lumped-model illustration excludes shaft charging and bearing-film breakdown; it is not a bearing-discharge prediction.

techniques [98]. In SPB, this mechanism acts at each of the N winding groups individually; because (14)–(15) gives each group a different absolute CM reference, the resulting bearing- and shaft-voltage stress is not necessarily uniform across the stack, and whether it is dominated by the highest-offset cell, by all cells combined, or by resonant interaction between cells remains open. Shaft-voltage cancellation circuits developed for conventional drives [99] are a plausible mitigation candidate but have not been demonstrated for a series-stacked, multi-port machine.

The same switching events also produce fast dv/dt transitions at the winding terminals, stressing turn-to-turn and phase-to-frame insulation independently of the CM current path. WBG devices, already motivated for SPB by their favorable scaling at reduced blocking voltage (Section VIII), shorten rise time and increase overshoot, and have been shown to accelerate insulation aging and partial-discharge activity in hairpin and other traction-machine windings [100]–[102]; diagnostic methods exist to assess the resulting insulation condition [103], within the broader set of insulation technologies for transport-electrification machines [104]. In SPB, this stress is superimposed on the stack-position-dependent offset of (14), so insulation coordination

must account for both the local switching transition and the cell’s absolute stack position, consistent with the challenge identified qualitatively in Section IV.

Mitigation therefore has to act at more than one level. Active EMI filtering can attenuate common-mode current at the converter terminals without relying on machine or bearing hardware [105], and cancellation circuits can target shaft voltage directly [99]. At the modulation level, Section V showed that zero-sequence injection and redundant-vector selection reduce CMV excitation [86], [87], but the carrier-phase displacement used there to interleave switching harmonics for ripple reduction, (8), is chosen to cancel a selected ripple harmonic and is not simultaneously optimized for common-mode excitation. Because SPB’s CM problem additionally depends on the fixed stack offset $V_{-,k}$, which carrier phasing cannot change, an interleaving choice that minimizes dc-link or stator ripple need not minimize aggregate common-mode excitation, and vice versa – a genuine SPB-specific design tension that future studies should address by extending the multi-objective modulation problem of Section V to common-mode and insulation-stress metrics evaluated across the full stack.

VII. DC-Link Dynamics, Stability, and Voltage Balancing

The common dc current and independently controlled winding-group powers (Section III) couple the local capacitor energies, so equal cell voltages are not inherently maintained [44], [45]. Related series-connected motor-drive research likewise shows that distributed current control alone does not ensure stable voltage sharing [89].

For cell k , the power balance is

$$C_k V_{dc,k} \frac{dV_{dc,k}}{dt} = V_{dc,k} i_{dc} - p_{ac,k} - p_{loss,k}. \quad (18)$$

Because the series current is common, dc input power depends on local voltage, whereas the winding-group controller determines ac output power. Defining the deviation from the mean cell voltage as

$$\tilde{V}_k = V_{dc,k} - \frac{1}{N} \sum_{j=1}^N V_{dc,j}. \quad (19)$$

a simplified differential-mode model around a balanced operating point is

$$CV_0 \frac{d\tilde{V}_k}{dt} \approx I_{dc,0} \tilde{V}_k - \Delta P_k. \quad (20)$$

where C and V_0 are the common cell capacitance and equilibrium voltage, $I_{dc,0}$ is the equilibrium dc current, and ΔP_k is the differential output-power perturbation. During motoring, $I_{dc,0} > 0$, a positive voltage perturbation increases cell input power and reinforces the deviation if output power remains unchanged. During regeneration, this feedback reverses. This interpretation agrees with SPB stability analyses [44], [45] and related series-connected multiphase-drive studies [106].

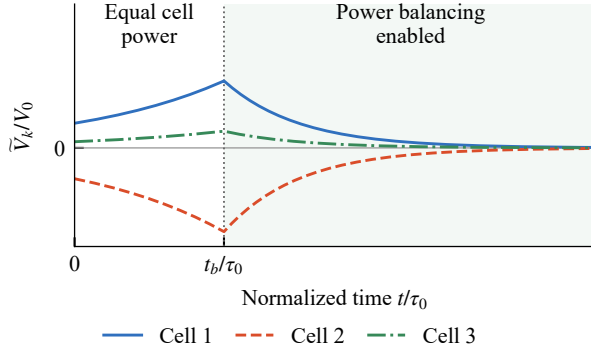


FIGURE 12. Conceptual analytical illustration of voltage balancing for three cells during motoring, based on (20) and [45]. Small deviations from the mean grow under equal constant cell powers. Enabling differential power balancing at t_b makes them decay toward zero while $\sum_k \Delta P_k = 0$. Time is normalized by $\tau_0 = CV_0/I_{dc,0}$; amplitudes and gain are illustrative, not a hardware transient or a prediction of settling time.

The constant-power-load model considered in [45] comprises

$$\begin{aligned} L_b \frac{di_{dc}}{dt} &= E_b - R_b i_{dc} - \sum_k V_{dc,k}, \\ C_k \frac{dV_{dc,k}}{dt} &= i_{dc} - \frac{P_k}{V_{dc,k}}. \end{aligned} \quad (21)$$

Here, E_b is the dc-source voltage, and L_b and R_b are the lumped series inductance and resistance of the shared source-to-stack supply path, including source and interconnection contributions; P_k is the cell's constant power demand. Under this model's assumptions, linearization about the open-loop motoring equilibrium for $N > 1$ yields $N - 1$ strictly positive real eigenvalues, establishing instability irrespective of parameter selection.

Fig. 12 illustrates the balancing principle using the small-signal model in (20), consistent with [45]. At equal constant output power, a lower-voltage cell draws more converter input current ($P_k/V_{dc,k}$) and its capacitor discharges further; a higher-voltage cell exhibits the opposite response. Thus, a small mismatch grows during motoring unless differential power is adjusted. The law $\Delta P_k = K_p \bar{V}_k$, motivated by controller alternative I in [45], increases output power in higher-voltage cells and decreases it in lower-voltage cells. For $K_p > I_{dc,0}$, (20) predicts exponential decay of the deviations. This local result assumes identical capacitances and ideal power-command tracking; it does not imply instability under every converter-machine control strategy.

The winding groups provide control authority over internal capacitor energy through differential power commands:

$$P_k^* = \frac{P^*}{N} + \Delta P_k, \quad \sum_{k=1}^N \Delta P_k = 0, \quad (22)$$

This allocation redistributes cell power without changing the total machine-power reference. Available balancing authority depends on phase-current margin, winding voltage, torque-ripple limits, and thermal state.

TABLE 4. Principal SPB dc-link balancing approaches and their main considerations.

Approach	Mechanism	Key consideration
Centralized	Differential current/power commands [44], [45]	Direct authority; global sensing and coordination
CDR	Common modulation, natural coupled dynamics [52], [55]	Low overhead; symmetry- and timing-sensitive
Distributed	Local agents, neighbor communication [53], [54], [56]	Scalable; requires communication and synchronization

Table 4 summarizes three principal approaches. Centralized control explicitly regulates voltage sharing through differential commands [44], [45]. Common-duty-ratio (CDR) control instead exploits the coupled converter-machine dynamics; stable sharing has been demonstrated for the investigated induction-machine system [52], with practical performance sensitive to timing mismatch [55]. Distributed multi-agent control extends local coordination to voltage balancing, power sharing, and reconfiguration [53], [54], [56]. Related modular-drive work uses local speed-droop laws to coordinate module power sharing [107].

For automotive applications, balancing must remain effective during torque transients, field weakening, parameter mismatch, and module reconfiguration. Sensor offsets, communication latency, propagation delays, and differential power limits determine the usable cell-voltage margin and therefore influence semiconductor voltage-rating selection. Dc-link stability must consequently be assessed across the coupled converter, machine, and control system.

VIII. Semiconductor Technology and Cell-Count Scaling

Increasing the SPB cell count N enables lower-voltage semiconductor devices, but voltage-rating selection must account for maximum blocking stress:

$$V_{\text{rated}} \geq k_{\text{rel}} \left(\frac{V_{\text{dc,max}}}{N} + \Delta V_{\text{imb}}^{\text{max}} + \Delta V_{\text{sw}}^{\text{max}} \right), \quad (23)$$

where $\Delta V_{\text{imb}}^{\text{max}}$ is the maximum cell-voltage imbalance, $\Delta V_{\text{sw}}^{\text{max}}$ is the commutation overshoot, and k_{rel} is the selected voltage-rating margin factor. Cell-count selection therefore depends on discrete commercial voltage classes and their usable operating margins.

SiC is well suited to high-voltage traction, while voltage partitioning can make lower-voltage GaN devices attractive through favorable conduction and switching characteristics [24], [35], [57]. For the illustrative 1200-V dc link in Fig. 13, two cells operate at a nominal 600 V each, leaving limited headroom for 650-V devices. Three cells reduce this voltage to 400 V, providing greater allowance for imbalance and switching transients. Further increases in N enable lower-voltage device classes but increase the number of active

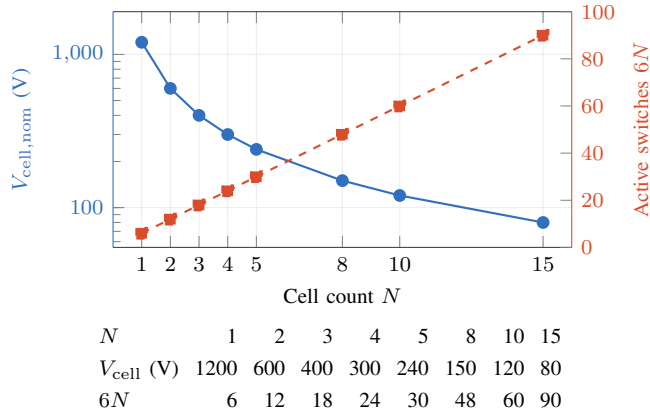


FIGURE 13. Cell-count scaling for a 1200-V dc link: nominal cell voltage $V_{\text{cell,nom}} = V_{\text{dc}}/N$ (blue circles, left logarithmic axis) and active switch positions $6N$ (red squares, right linear axis). The table gives the values at each design point. Illustrative device classes include 1700–2000-V SiC for $N = 1$, 650-V GaN/SiC for $N = 3$, and 100-V GaN or silicon MOSFETs for $N = 15$, subject to the required blocking-voltage margin. Device parallelization is excluded.

switch positions as $6N$. Final device selection must satisfy (23).

A. Commercial-Device Screening

Lower voltage ratings do not guarantee superior converter performance. Candidate devices must be assessed using conduction and switching characteristics together with current capability, thermal limits, and packaging constraints. Figures of merit such as $R_{\text{DS(on)}}C_{\text{oss}}$ and $R_{\text{DS(on)}}Q_g$ provide initial screening, but detailed comparison requires $R_{\text{DS(on)}}(T)$ for conduction loss and die-area requirements; $E_{\text{oss}}(V)$ for stored output-capacitance energy and its contribution to switching loss; $E_{\text{on/off}}(V, I, T)$ for switching-energy characterization; Q_g and Q_{gd} for gate-drive requirements and switching behavior; I_{cont} and R_{θ} for current derating, parallelization, and cooling; and package and power-loop inductance L_{σ} for commutation overshoot.

Machine phase current does not generally decrease as $1/N$. For a derated device current capability $I_{\text{dev,der}}$, the required parallel-device count and total active-device count can be estimated as

$$n_p = \left\lceil \frac{I_{\text{sw,max}}}{I_{\text{dev,der}}} \right\rceil, \quad N_{\text{sw,tot}} = 6Nn_p. \quad (24)$$

Consequently, favorable device-level figures of merit may coexist with substantial die area, package count, and gate-drive and thermal-interface requirements. High-current GaN modules illustrate the importance of low-inductance packaging and integrated gate driving at traction-relevant currents [108], [109].

Device selection and layout are further coupled through commutation overshoot:

$$\Delta V_{\text{sw}} \approx L_{\sigma} \frac{di}{dt}, \quad (25)$$

Package and interconnection inductance can consume much of the voltage margin gained through partitioning. High

switching speeds therefore require careful commutation-loop design, local decoupling, and gate-drive layout, particularly for GaN [109]. Dynamic $R_{\text{DS(on)}}$, reverse conduction, gate reliability, short-circuit capability, and thermal performance must also be considered [110], [111]. Room-temperature static resistance alone is therefore insufficient for cell-count selection.

B. Loss Scaling With Cell Count

For an individual device, conduction and switching losses can be approximated as

$$P_{\text{cond}} \approx I_{\text{rms,sw}}^2 R_{\text{DS(on)}}(T_j), \quad (26)$$

and

$$P_{\text{sw}} \approx f_{\text{sw}} [E_{\text{on}}(V, I, T_j) + E_{\text{off}}(V, I, T_j)]. \quad (27)$$

Including dead-time-related losses and gate-drive power gives

$$P_{\text{semi}}(N) = \sum_{k=1}^N \sum_{j=1}^{6n_p(N)} (P_{\text{cond},k,j} + P_{\text{sw},k,j} + P_{\text{dt},k,j}) + P_{\text{gate}}(N), \quad (28)$$

where $P_{\text{dt},k,j}$ represents dead-time-related device loss and P_{gate} is the aggregate gate-drive power.

Total loss does not scale linearly with N : additional cells increase device count, while transitions between voltage classes change on-state resistance, switching energy, thermal impedance, and parallelization requirements. Low-cell-count designs may favor high-voltage SiC, whereas lower-voltage GaN becomes attractive when its device, packaging, or passive-component benefits offset the additional modules.

Cell count and semiconductor technology must therefore be selected jointly using the voltage and component-count relationships in Fig. 13, commercial-device characteristics, and the machine requirements of Section IV. No single device figure of merit establishes the preferred SPB architecture.

IX. Fault Tolerance and Reconfiguration

SPB modularity offers potential graceful degradation because the converter and machine are divided into repeated converter–winding modules. However, fault tolerance is not automatic: because the dc ports are connected in series, a failed cell can disturb or interrupt the complete dc current path, a structural challenge demonstrated for SPB and for other series-connected modular converters alike [112]–[114]. Early SPB work demonstrated post-fault operation under semiconductor faults, establishing that repeated hardware must be accompanied by appropriate safe states and control action [112]. At the machine level, a submodule short-circuit fault in a modular converter feeding fractional-slot concentrated-winding PM machines was shown to drive a fault current above rated current because of the strong NdFeB rotor magnetization, risking machine overheating; reducing rotor magnetization limits this fault current but also

narrows the constant-power-speed range, a design tradeoff specific to fault-tolerant FSCW PM machines [113]. In the SPB machine designs of that study, post-fault short-circuit current with NdFeB magnets exceeds the healthy-phase current, whereas with ferrite magnets it stays below it, at the cost of becoming visibly non-sinusoidal from magnetic coupling with the adjacent healthy windings – a concrete illustration that PM material selection is itself a fault-tolerance design variable, not only a torque-density one. Fault handling in other series-connected modular converters further underscores that mitigation requires the control system to be explicitly designed for module isolation [114].

The machine provides a second layer of redundancy. Healthy winding groups can potentially redistribute current after phase or converter faults, as demonstrated in recent multiphase and modular-drive studies [62], [115], [116]. In SPB, however, this machine-side redundancy is useful only if the failed converter cell can be isolated or bypassed without breaking the series dc path.

Building on the online reconfiguration capability introduced in Section II [56], after a faulted module is isolated the remaining active set $\mathcal{A}$ must satisfy

$$P^* = \sum_{k \in \mathcal{A}} P_k^*, \quad (29)$$

with torque derating if current, voltage, or thermal limits prevent full redistribution. Reconfiguration can therefore provide both fault isolation and an operating-range degree of freedom because the number of active cells changes the voltage available to the remaining winding groups. Bypassing a cell raises the steady voltage of each surviving cell when the total bus voltage remains fixed. With $M = |\mathcal{A}|$ active cells, a necessary blocking-voltage condition is

$$\frac{V_{dc,max}}{M} + \Delta V_{imb} + \Delta V_{sw} \leq V_{allow}, \quad (30)$$

where V_{allow} is the allowable device voltage after the chosen design margin, and the two increments cover cell imbalance and switching overshoot. Capacitor and insulation limits must also hold. Current or torque derating cannot by itself correct excessive steady blocking voltage. If this condition fails, operation requires a lower bus voltage, a suitable spare-cell arrangement, or shutdown. Fig. 15 separates voltage feasibility from post-fault torque capability. Fig. 14 maps these fault locations onto the SPB structure.

Reliability remains ambiguous because increasing cell count provides more potential degraded states but also more components that can fail. System or mission reliability is therefore more meaningful than component count alone. Markov-chain analysis has been applied to related fault-tolerant IMMDs to distinguish healthy, degraded, and failed states [118], while integrated distributed traction drives provide additional evidence that fault tolerance must include both the machine and the local converter architecture [70], [71].

The direct SPB literature therefore demonstrates the feasibility of post-fault operation and online reconfiguration, but

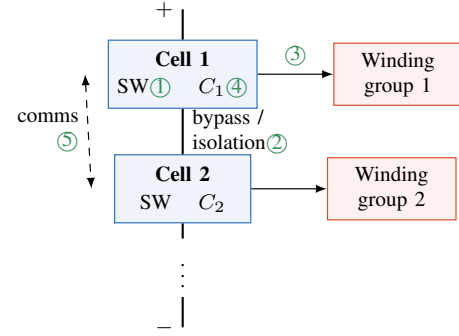


FIGURE 14. Simplified SPB converter-machine structure with fault locations ①–⑤ annotated, corresponding in order to the rows of Table 5.

TABLE 5. SPB Fault Locations and Required System Response

Fault	SPB consequence	Required response
Switch / bridge	Cell cannot synthesize commanded voltage	Fast safe state and protection [112]
Converter module	Series dc path can be interrupted	Bypass/isolation and power redistribution [56]
Winding phase	Reduced and asymmetric torque capability	Multiphase post-fault current control [62], [116]
Capacitor sensor*	Loss of voltage-sharing information or margin	Detection, derating, isolation or shutdown
Communication / controller	Loss of distributed coordination	Local fallback and deterministic safe state [117]

*The SPB studies surveyed here do not establish capacitor or sensor fault coverage; this row reflects the authors' system-level inference from the general modular-converter requirement to detect loss of cell-voltage sensing or local dc-link capacitance and to derate or isolate the affected cell accordingly.

not yet automotive-level fault coverage. Detection and isolation time, short-circuit energy, capacitor and sensor faults, communication loss, post-fault thermal loading, diagnostic coverage, and deterministic functional-safety transitions remain important validation topics.

X. Research Gaps and Future Directions

SPB competitiveness depends on joint battery, converter, and machine design. Voltage partitioning, winding constraints, modulation, balancing, common-mode excitation, and fault recovery must be evaluated together; improving one subsystem can increase losses or complexity elsewhere. The following framework translates these dependencies into application requirements, followed by priorities for validation.

A. Application-Based Design and Validation Boundary

A useful co-design starts from the battery voltage range, required shaft torque-speed trajectory, cooling boundary, and permitted post-fault operation. Let the candidate design be

$$\mathbf{x} = \{N, \mathcal{D}, n_{||}, \mathcal{W}, C_k, f_{sw}, \mathcal{M}, \mathcal{H}\}, \quad (31)$$

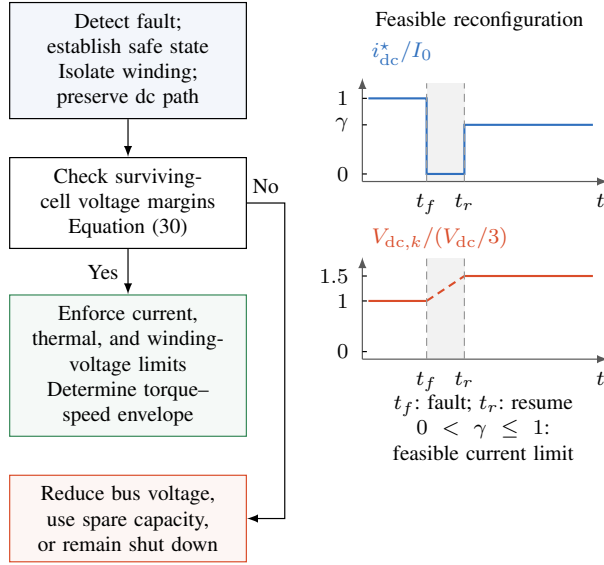


FIGURE 15. Post-fault feasibility assessment with illustrative waveforms for the feasible branch. The dc-current command is temporarily set to zero and restored to γI_0 , subject to operating limits. For a fixed bus, changing from three to two active cells raises the balanced surviving-cell voltage from 1 to 1.5 p.u. (400 to 600 V for a 1200-V bus). Shading marks the protection/reconfiguration interval; the dashed voltage connection indicates a change of steady-state level, not a predicted capacitor-voltage trajectory. Current derating does not remove the increased blocking-voltage requirement.

where $\mathcal{D}$, $\mathcal{W}$, $\mathcal{M}$, and $\mathcal{H}$ denote the device, winding, modulation/control, and cooling choices, and $n_{||}$ is the number of parallel devices per switch. The proposed comparison minimizes the objective vector

$$\mathbf{J}(\mathbf{x}) = [E_{\text{mission}}, \mathcal{V}_{\text{drive}}, m_{\text{drive}}, C_{\text{life}}], \quad (32)$$

subject to healthy and specified degraded-state constraints on device voltage, winding voltage/current, junction and winding temperature, capacitor ripple, insulation, and common-mode excitation. Mission energy is the signed battery-terminal energy $\int V_b i_b dt$ for the same shaft duty, including regeneration and drive/cooling auxiliaries within the declared boundary. Reliability is a service requirement or an additional objective when comparable failure data are available. These equations define a proposed evaluation framework, not a completed optimization or a universal optimum cell count.

A fair benchmark compares SPB with two-level and appropriate three-level drives at the same battery envelope, shaft duty, coolant conditions, EMC limits, and fault requirements. Each architecture should be optimized within those constraints. Report inverter-only and complete-drive volume separately, including capacitors, filters, isolation supplies, connectors, protection, and cooling. Use temperature-dependent device loss data and identify uncertainty in parasitics, mission weighting, and lifetime assumptions. A nominal-point efficiency gain is insufficient if it disappears after auxiliary loss, field weakening, or added machine losses are included.

Table 6 connects representative direct SPB results to the remaining EV validation. The review does not imply that related IMMD results close these gaps. SPB may be unattractive when a suitable multiphase machine is unavailable, device savings are small, interconnection/insulation overhead dominates, or the required post-fault voltage margin forces the same device class as a simpler architecture. Reporting these boundaries is as important as identifying favorable designs.

B. Research Priorities

The optimum cell count N remains the central unresolved design question: as established in Section III, the voltage-versus-component-count tradeoff shifts the optimum as SiC, GaN, packaging, and cooling evolve [3], [24]. A common-platform comparison of a few-cell 650-V GaN SPB against a many-cell low-voltage GaN SPB, using real commercial devices, would be valuable; the GaN comparison in [57] is a useful starting point but should be extended to automotive power with current parallelization, machine voltage/speed constraints, dc-link capacitance, cooling, and mission energy.

Converter, machine, and thermal models, among others, should be co-optimized rather than designed sequentially. Existing phase-count studies show machine design changes with SPB cell number [33]; the missing step is a complete automotive co-design using vehicle-level energy, volume, mass, cost, and reliability as a joint objective.

Reliability is part of that objective: future studies should combine semiconductor, capacitor, sensor, gate-driver, communication, and winding faults (Section IX; [56], [112]) in one state-based, Markov-style framework as used for related fault-tolerant IMMDs [118], keeping *component failure*, *module isolation*, and *loss of vehicle function* distinct. An N -cell drive may survive with $N - 1$ active modules, but its torque-speed envelope, module thermal loading, and dc-voltage margin all change, so reliability data should pair with a post-fault capability model – not a series reliability block diagram – to compare availability, limp-home power, and mission completion probability against efficiency.

Cost and sustainability assessment should use the same N -dependent framing: increasing N can reduce per-switch die cost but tends to increase isolated gate supplies, sensing channels, and assembly operations, among others, almost linearly, while a repeated converter-winding module can gain from common tooling and automated assembly – a balance likely to differ between high-volume passenger vehicles and lower-volume heavy-duty systems. The same accounting should separate production, operation, maintenance, and end-of-life impact over one vehicle mission and service life, since modular repair and longer service life can offset the added production impact of extra components.

Overall, SPB research must move from demonstrating that the architecture operates toward demonstrating where it is better: whether it gives a measurable vehicle-level advantage over mature two- and three-level architectures

TABLE 6. Direct SPB evidence and remaining EV validation: synthesis of representative studies

Direct SPB literature	Established focus	Additional EV validation needed
Control and modulation [42], [46]	Coordinated switching and dc-current harmonic shaping	Mission-wide loss/ripple tradeoff with measured inter-connect parasitics
Machine design [33]	Cell/phase count coupled to converter–machine design	Common battery, cooling, insulation, and packaging boundary
Balancing [52]–[54]	Common-duty-ratio, distributed balancing, and power-sharing methods	Control authority during field weakening, regeneration, saturation, and faults
GaN comparison [57]	Series-stacked two-level versus flying-capacitor implementation	Traction-scale comparison including filters, winding insulation, and cooling
Fault operation [56], [112]	Post-fault operation and module reconfiguration	Verified isolation, surviving-cell voltage margins, and degraded torque–speed envelope

once machine design, thermal management, EMC, reliability, manufacturing, and cost are included.

XI. Conclusion

This paper reviewed stacked polyphase bridge (SPB) converters – series-stacked two-level bridges each feeding an electrically separated winding group – as a candidate architecture for future high-voltage traction drives, partitioning the dc-link voltage while introducing converter–machine modularity, distributed control, and fault-management opportunities.

The review also shows that the semiconductor voltage advantage cannot be separated from the complete traction system. Increasing cell count adds converter, control, and insulation overhead, so SPB does not eliminate the complexity of high-voltage conversion but redistributes it between semiconductor technology, machine architecture, and control. WBG development makes this trade increasingly relevant, particularly where SPB can move a high-voltage traction system toward lower-voltage GaN devices; whether that approach is advantageous depends on whether the device-level gains compensate for the additional modular hardware and machine requirements.

The remaining priority is integrated co-design across the battery/dc-link, semiconductor/converter, and machine sides, evaluated jointly against vehicle-level energy, cost, and reliability targets rather than compared through separate, narrower studies. The evidence-to-validation synthesis and proposed design framework make the evaluation boundary explicit. SPB is a modular traction-drive architecture that exchanges semiconductor blocking-voltage stress for converter–machine modularity. Its competitiveness depends on whether that exchange produces a measurable vehicle-level benefit.

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